

Lab 3: Bash Scripting

1. Commands:

```
ls
mkdir bash
ls
cd bash
nano basic.sh
```

Write text using 'echo' in nano text editor:

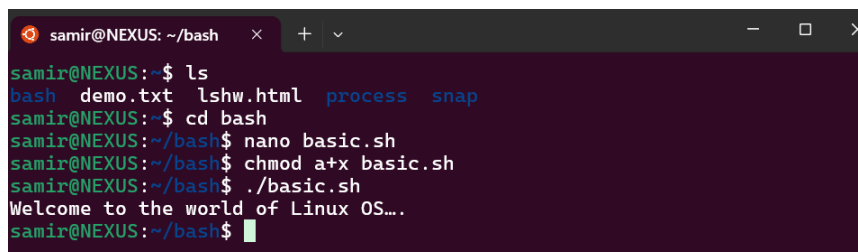
```
echo "Welcome to the world of Linux OS..."
```

Press ctrl+x to save file.

```
chmod a+x basic.sh
./basic.sh
```

Interpretation: ls to list files/directories, mkdir to make directory, cd to change directory, nano to make file writable, echo to write in file.

Output:

A terminal window titled 'samir@NEXUS: ~/bash' showing the following commands and output: 'ls' lists 'bash demo.txt lshw.html process snap'; 'cd bash' changes the directory; 'nano basic.sh' creates the file; 'chmod a+x basic.sh' makes it executable; and './basic.sh' outputs 'Welcome to the world of Linux OS...'.

```
samir@NEXUS: ~/bash
samir@NEXUS:~$ ls
bash demo.txt lshw.html process snap
samir@NEXUS:~$ cd bash
samir@NEXUS:~/bash$ nano basic.sh
samir@NEXUS:~/bash$ chmod a+x basic.sh
samir@NEXUS:~/bash$ ./basic.sh
Welcome to the world of Linux OS...
samir@NEXUS:~/bash$
```

2. Commands:

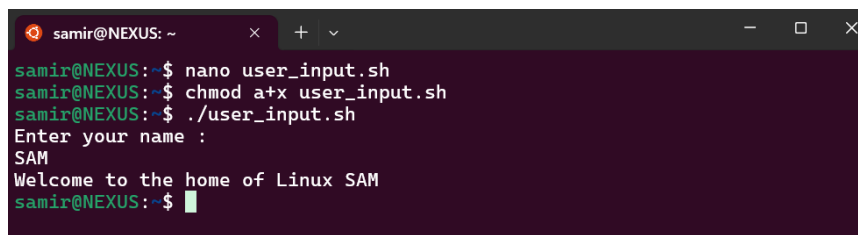
```
nano user_input.sh
```

Write text in nano:

```
echo "Enter your name : "
read name
echo "Welcome to the home of Linux $name"
```

Interpretation: Takes input from user and shows output.

Output:

A terminal window titled 'samir@NEXUS: ~' showing the following commands and output: 'nano user_input.sh' creates the file; 'chmod a+x user_input.sh' makes it executable; './user_input.sh' prompts for a name, which is 'SAM', and then outputs 'Welcome to the home of Linux SAM'.

```
samir@NEXUS: ~
samir@NEXUS:~$ nano user_input.sh
samir@NEXUS:~$ chmod a+x user_input.sh
samir@NEXUS:~$ ./user_input.sh
Enter your name :
SAM
Welcome to the home of Linux SAM
samir@NEXUS:~$
```

3. Commands:

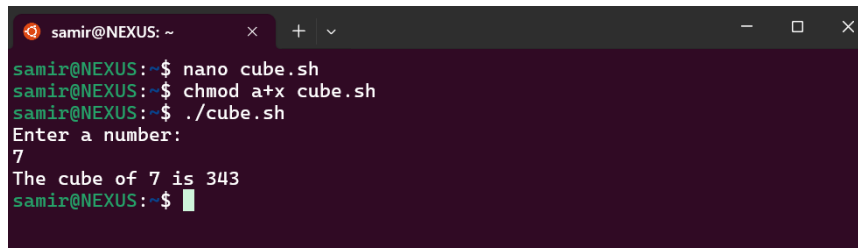
```
nano cube.sh
```

Write in nano:

```
echo "Enter a number: "  
read number  
cube=$(( $number*$number*$number ))  
echo "The cube of $number is $cube"
```

Interpretation: Takes input number and calculates its cube.

Output:



```
samir@NEXUS: ~  
samir@NEXUS:~$ nano cube.sh  
samir@NEXUS:~$ chmod a+x cube.sh  
samir@NEXUS:~$ ./cube.sh  
Enter a number:  
7  
The cube of 7 is 343  
samir@NEXUS:~$
```

4. Commands:

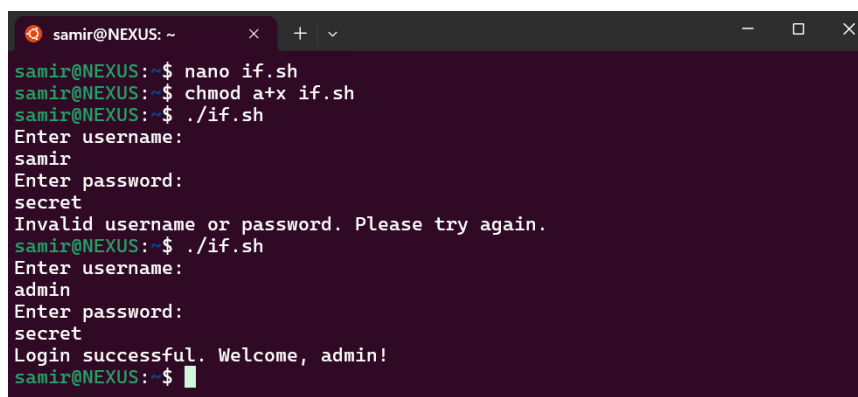
```
nano if.sh
```

Write in nano:

```
echo "Enter username: "  
read username  
echo "Enter password: "  
read password  
if [[ ($username == "admin" && $password == "secret") ]]; then  
echo "Login successful. Welcome, $username!"  
else  
echo "Invalid username or password. Please try again."  
fi
```

Interpretation: Validates username and password input.

Output:



```
samir@NEXUS: ~  
samir@NEXUS:~$ nano if.sh  
samir@NEXUS:~$ chmod a+x if.sh  
samir@NEXUS:~$ ./if.sh  
Enter username:  
samir  
Enter password:  
secret  
Invalid username or password. Please try again.  
samir@NEXUS:~$ ./if.sh  
Enter username:  
admin  
Enter password:  
secret  
Login successful. Welcome, admin!  
samir@NEXUS:~$
```

5. Commands:

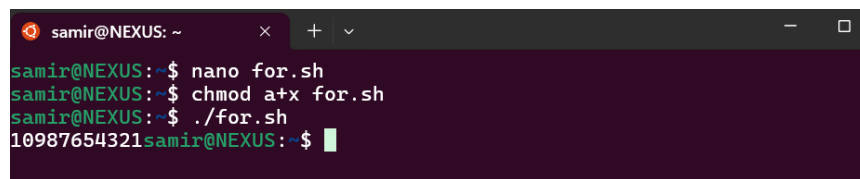
```
nano for.sh
```

Write in nano:

```
#!/bin/bash
for((counter=10;counter>0;counter--))
do
echo -n "$counter"
done
```

Interpretation: Prints numbers from 10 to 1 using for loop.

Output:

A terminal window titled 'samir@NEXUS: ~' showing the execution of a script. The user runs 'nano for.sh', then 'chmod a+x for.sh', and finally './for.sh'. The output is '10987654321' followed by the prompt 'samir@NEXUS:~\$'.

6. Commands:

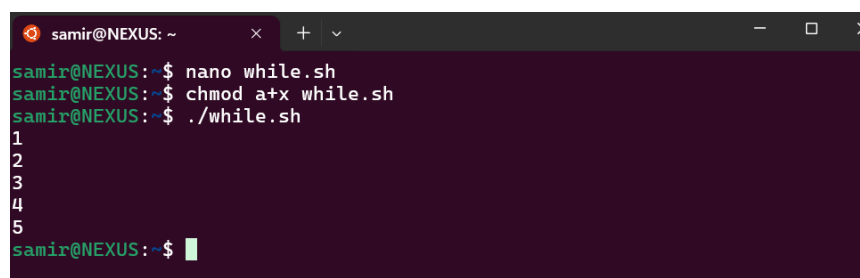
```
nano while.sh
```

Write in nano:

```
#!/bin/bash
valid=true
count=1
while [ $valid ]
do
echo $count
if [ $count -eq 5 ];
then
break
fi
((count++))
done
```

Interpretation: Prints numbers from 1 to 5 using while loop.

Output:

A terminal window titled 'samir@NEXUS: ~' showing the execution of a script. The user runs 'nano while.sh', then 'chmod a+x while.sh', and finally './while.sh'. The output is '1', '2', '3', '4', '5' on separate lines, followed by the prompt 'samir@NEXUS:~\$'.

7. Commands:

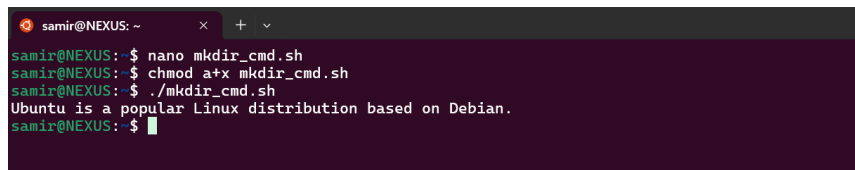
```
nano mkdir_cmd.sh
```

Write in nano:

```
#!/bin/bash
mkdir ubuntu
cd ubuntu
touch ubuntu.txt
echo "Ubuntu is a popular Linux distribution based on Debian.">ubuntu.txt
cat ubuntu.txt
```

Interpretation: Creates ubuntu directory, makes ubuntu.txt, adds content, displays file.

Output:



```
samir@NEXUS: ~
samir@NEXUS:~$ nano mkdir_cmd.sh
samir@NEXUS:~$ chmod a+x mkdir_cmd.sh
samir@NEXUS:~$ ./mkdir_cmd.sh
Ubuntu is a popular Linux distribution based on Debian.
samir@NEXUS:~$
```

8. Commands:

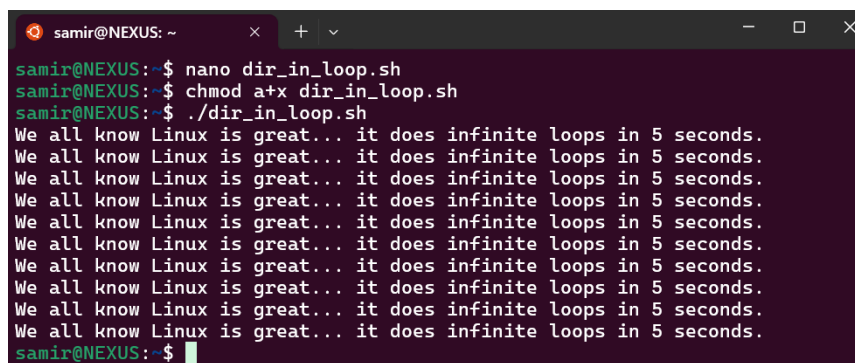
```
nano dir_in_loop.sh
```

Write in nano:

```
#!/bin/bash
num_iterations=10
mkdir this
cd this
for (( i=1; i<=$num_iterations; i++ ))
do
touch this_$i.txt
echo "We all know Linux is great... it does infinite loops in 5 seconds." > this_$i.tx
t
cat this_$i.txt
done
```

Interpretation: Creates 'this' directory, generates 10 text files with same message.

Output:



```
samir@NEXUS: ~
samir@NEXUS:~$ nano dir_in_loop.sh
samir@NEXUS:~$ chmod a+x dir_in_loop.sh
samir@NEXUS:~$ ./dir_in_loop.sh
We all know Linux is great... it does infinite loops in 5 seconds.
We all know Linux is great... it does infinite loops in 5 seconds.
We all know Linux is great... it does infinite loops in 5 seconds.
We all know Linux is great... it does infinite loops in 5 seconds.
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We all know Linux is great... it does infinite loops in 5 seconds.
We all know Linux is great... it does infinite loops in 5 seconds.
We all know Linux is great... it does infinite loops in 5 seconds.
samir@NEXUS:~$
```

9. Commands:

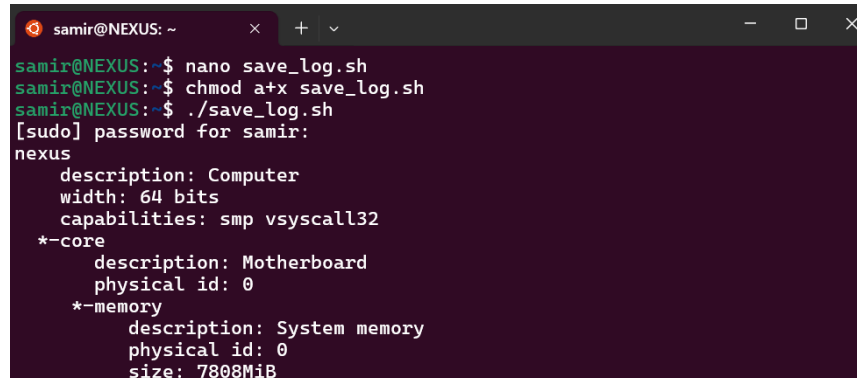
```
nano save_log.sh
```

Write in nano:

```
sudo lshw
sudo lshw
sudo lshw -html > lshw.html
```

Interpretation: Runs listed commands.

Output:

A terminal window titled 'samir@NEXUS: ~' showing the execution of a script. The user runs 'nano save_log.sh', then 'chmod a+x save_log.sh', and finally './save_log.sh'. The script outputs the system information from 'lshw', including details about the computer, motherboard, and system memory.

```
samir@NEXUS: ~  
samir@NEXUS:~$ nano save_log.sh  
samir@NEXUS:~$ chmod a+x save_log.sh  
samir@NEXUS:~$ ./save_log.sh  
[sudo] password for samir:  
nexus  
description: Computer  
width: 64 bits  
capabilities: smp vsyscall32  
*-core  
description: Motherboard  
physical id: 0  
*-memory  
description: System memory  
physical id: 0  
size: 7808MiB
```

10. Commands:

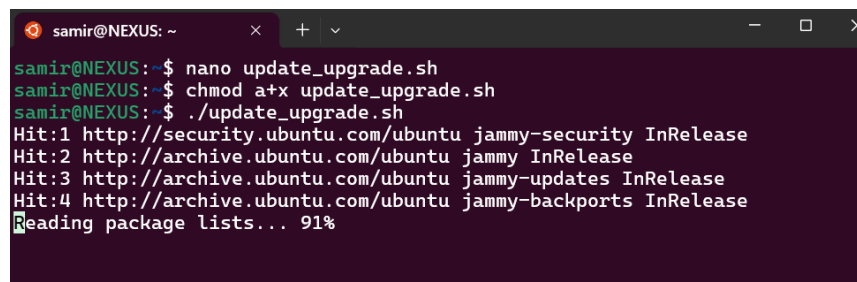
```
nano update_upgrade.sh
```

Write in nano:

```
sudo apt-get update  
echo "Updated"  
sudo apt-get upgrade  
echo "Upgraded"
```

Interpretation: Performs system update and upgrade.

Output:

A terminal window titled 'samir@NEXUS: ~' showing the execution of a script. The user runs 'nano update_upgrade.sh', then 'chmod a+x update_upgrade.sh', and finally './update_upgrade.sh'. The script outputs the results of 'apt-get update' and 'apt-get upgrade', showing several security updates for the jammy distribution.

```
samir@NEXUS: ~  
samir@NEXUS:~$ nano update_upgrade.sh  
samir@NEXUS:~$ chmod a+x update_upgrade.sh  
samir@NEXUS:~$ ./update_upgrade.sh  
Hit:1 http://security.ubuntu.com/ubuntu jammy-security InRelease  
Hit:2 http://archive.ubuntu.com/ubuntu jammy InRelease  
Hit:3 http://archive.ubuntu.com/ubuntu jammy-updates InRelease  
Hit:4 http://archive.ubuntu.com/ubuntu jammy-backports InRelease  
Reading package lists... 91%
```